

■ Stock Market Forecasting Dashboard — Project Summary

A simplified, structured explanation of every component

1. What Is This Project?

This project builds an **end-to-end Stock Market Forecasting Dashboard**. It automatically collects real stock data, cleans it, trains AI models to spot patterns, predicts future prices, and shows everything in a live, interactive website anyone can use.



Core Goal

Collect historical stock data → Clean it → Train forecasting models → Predict future prices → Show in a dashboard → Host on cloud.

2. How Data Is Collected

The project uses a Python library called **yfinance** to fetch live and historical stock data directly from **Yahoo Finance** — completely free and automatic.

Field	What It Means
Open	Price at which the stock starts trading for the day
Close	Price at which the stock stops trading (most important)
High	Highest price reached during the day
Low	Lowest price reached during the day
Volume	Total number of shares traded that day
Date	The calendar day this data belongs to

3. Data Preprocessing (Cleaning)

Raw data downloaded from the internet is messy and incomplete. Preprocessing fixes all of that before any AI model sees the data.

A	Remove Missing Values Some days may have empty or incomplete records. These are deleted so the model only trains on reliable data.
B	Fix the Date Format Dates must be in a standard format (YYYY-MM-DD) so the model understands the correct time order.
C	Set Date as the Index Time-series models work by marching through time. Setting the date as the timeline allows the model to look backwards correctly.
D	Select the Close Price Out of all daily fields, the Closing Price is the most meaningful — it reflects the final agreed market value of the stock.
E	Scale Values 0–1 (LSTM only) Neural networks work best when numbers are tiny (0 to 1). A scaler shrinks all prices into that range, then expands predictions back after.

4. Forecasting Models

Four different AI / statistical models are used to predict future stock prices. Each has its own strengths — comparing them reveals which one works best.

■	ARIMA — Auto-Regressive Integrated Moving Average A classic mathematical model that studies past price patterns and finds hidden trends. Think of it like a careful accountant reviewing your spending history to predict next month's expenses. Best for: Simple, steady trends with no seasonality
■	SARIMA — Seasonal ARIMA ARIMA's smarter sibling. It does everything ARIMA does PLUS recognises repeating seasonal cycles — e.g. stocks rising every month-end or every quarter. Best for: Data with regular seasonal patterns
■	Prophet — Developed by Meta (Facebook) A ready-made business forecasting tool. You hand it data; it automatically finds trends, seasonality, and holiday effects — no math knowledge needed. Best for: Business data, holiday/seasonal effects



LSTM — Long Short-Term Memory Neural Network

A deep-learning model that works like human long-term memory. It can remember patterns from months ago and use them alongside recent data to make more nuanced predictions.

Best for: Complex, nonlinear, long-range patterns

5. How Accuracy Is Measured

After a model makes predictions, we compare them against the actual stock prices to score the model's accuracy. Two main metrics are used:

MAE

Mean Absolute Error

The average gap between predicted price and real price. E.g. if MAE = ₹5, the model is off by ₹5 on average.

■ Lower = Better

RMSE

Root Mean Square Error

Similar to MAE but punishes large errors more heavily. A low RMSE means few big surprises.

■ Lower = Better

6. The Interactive Dashboard

The dashboard is built with **Streamlit** — a Python tool that turns your code into a live website instantly. No web-development experience needed.

Icon	Feature	What It Does
■ ■	Pick Any Stock	Choose TCS, Reliance, Infosys, or any ticker symbol.
■	View Price Graphs	Interactive charts showing historical price movements.
■	Compare All Models	See predictions from ARIMA, SARIMA, Prophet, and LSTM side by side.
■	Accuracy Scores	MAE and RMSE shown for each model so you know which to trust.
■	Future Price Outlook	A forward-looking chart showing predicted prices for coming days.

7. Cloud Deployment on AWS

The dashboard is hosted on **Amazon EC2** — a virtual computer in the cloud. This means anyone with the link can open the dashboard from anywhere in the world, 24 hours a day, without installing anything.

1	Upload	Project code is uploaded to the EC2 server
2	Setup	Python environment and all libraries are installed on the server
3	Run	Streamlit starts the web app on the server
4	Access	Users open the dashboard in any browser via a URL

8. Complete Project Workflow

■	Yahoo Finance	Source of all stock data
■ ■	yfinance Library	Python tool that downloads the data
■	Preprocessing	Clean, format, and scale the data
■	Train 4 Models	ARIMA · SARIMA · Prophet · LSTM
■	Evaluate Accuracy	Measure MAE and RMSE for each model
■	Generate Predictions	Forecast future stock prices
■ ■	Streamlit Dashboard	Display charts and predictions interactively
■ ■	AWS EC2	Host the dashboard for public access